

Lab 5 - Two Sum

Course: CS101 Introduction to Programming

Language: Python 3

Topic: Arrays & hashing

Task

Given a list of integers `nums` and an integer `target`, return the indices of the two numbers that add up to `target`:

```
def two_sum(nums: list[int], target: int) -> list[int]:  
    ...
```

- Return a list of the two indices `[i, j]` (order does not matter).
- You may assume exactly one valid pair exists.
- You may not use the same element twice (the two indices must be different).

Example: `two_sum([2, 7, 11, 15], 9)` returns `[0, 1]` because `nums[0] + nums[1] == 9`.

What we assess

1. Correctness - returns the indices of a valid pair on all test inputs.
2. Handling of tricky inputs - duplicate values, negative numbers, the pair at the ends.
3. Returns indices, not the values, and never reuses one element.